Magnetic fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret magnetic field patterns and compass direction.

SUCCESS CRITERIA - I can...

1. Label north and south poles.
2. Trace external field direction from north toward south.
3. Explain why field-line drawings are models.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 5 Ω resistor is in series with two parallel 6 Ω resistors across 16 V. Find parallel equivalent, total resistance and supply current.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $B_{net,\,x} = B_{1,\,x} + B_{2,\,x}$; $B_{net,\,y} = B_{1,\,y} + B_{2,\,y}$. Add field components at the same point. A compass follows the net field. Outside a bar magnet, arrows run north to south.

Worked reasoning: Outside a bar magnet, field arrows run from the north pole toward the south pole. A compass's north-seeking end follows the local field direction.

Your turn: Draw curved external field lines and two small compass arrows at different points.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Label N and S on the bar magnet. Sketch external field arrows and the direction a nearby compass would point.



Marzano check - circle: ① ② ③

04 Apply the physics

A compass between two identical magnets has no stable preferred direction at one location. Draw two possible opposing field contributions there. Explain how superposition could account for the observation and why it does not prove every field is zero.

● Field / first contribution

● Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student draws field lines that cross at one point. Explain why this is inconsistent with a single net field direction at that point.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

In separate cases, an ideal blackbody temperature becomes 4 times as large and a sound frequency becomes 6 times as large at fixed sound speed. Compare the two wavelength changes.
Keep the two relationships separate and label each ratio.

Case	Quantity changed	Wavelength ratio
Thermal spectrum	Temperature	___
Sound at fixed speed	Frequency	___

Marzano check - circle: 1 2 3

07 Exit check

If a bar magnet is cut in half, do you obtain an isolated north pole and an isolated south pole? Explain the usual result.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two magnets face N toward N. Sketch the field qualitatively and predict the force. Then reverse one magnet and explain both changes.

Label poles; add force or field arrows.



Marzano check - circle: 1 2 3